Run report: cover census, run 20261013T0305Z

Descriptive record of one run. Interpretation is in the study `README.md` / `REPORT.pdf`.

What ran

- Command: `rustc -O cover_census.rs && ./cover_census 5 && ./cover_census 7 && ./cover_census 11 120`, 8 threads.
- Spec: `PROTOCOL.md`, frozen before the run. Fixed seed 20261012 for the q = 11 sample.
- Isolation: none; shared host under heavy load. No timing is a result. All recorded values are exact counts and deterministic.
- Raw records: `results/run-20261013T0305Z.jsonl` (`rec` = field, curve, summary, totals).

```
run: 20261013T0305Z
command: rustc -O cover_census.rs && ./cover_census 5 && ./cover_census 7 && ./cover_census 11 120   (8
threads)
commit (base): ecb69d15aeb2eba2326ffa6f6fe4e2737efb05d1
source sha256: d11e5678094801eefb4d55d890cefdf314023631da01352842fe1bc7a3868115
rustc: rustc 1.93.1 (01f6ddf75 2026-02-11) (Homebrew)
host: Darwin 25.6.0 arm64, Apple M4 Pro, 14 cpus
uptime at start: 20:05  up 1 day, 19:44, 11 users, load averages: 89.41 99.77 105.16
q=5 real 1.87
q=7 real 37.57
note: the q=11 arm first failed on a shell quoting error (arguments passed as one string, no output); rerun
below with the same binary
q=11 120 real 203.46
uptime at end: 20:09  up 1 day, 19:48, 11 users, load averages: 128.58 119.01 112.52
```

Runs requested and completed

Arm	Requested	Completed	Notes
q = 5	all classes, all structures	252 curves x 16275 structures	
q = 7	all classes, all structures	690 curves x 120099 structures	
q = 11	120 sampled curves and their 2-isogeny neighbours, all structures	120 curves x 1786323 structures	first attempt failed on a shell quoting error and produced no output; rerun with the same binary

Recorded values

Unit: number of isomorphism classes of curves (q = 5, 7) or sampled curves (q = 11). Counts are exact; no uncertainty is attached. `mismatches` (genus formula against the sum over quadratic subcovers) is 0 in all three arms.

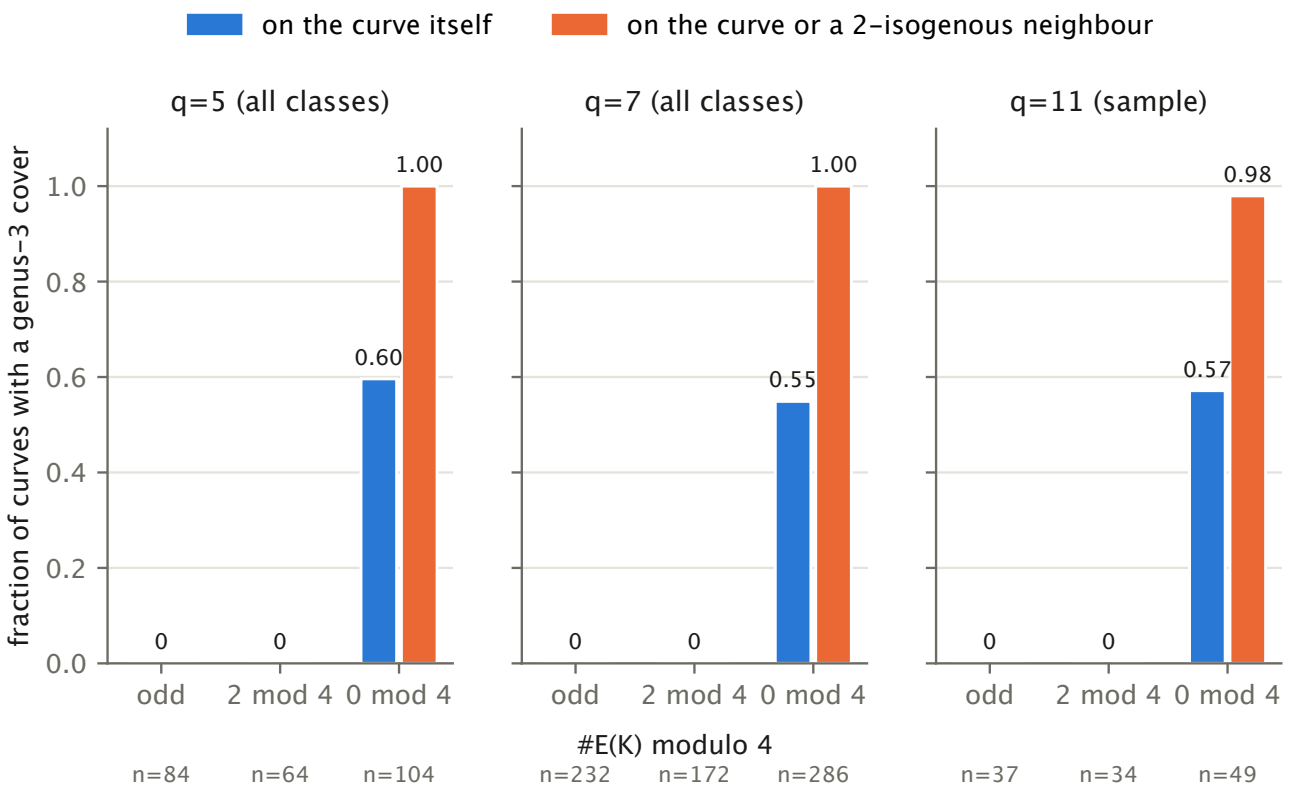


Figure 1: Run 20261013T0305Z. Fraction of curves whose minimal cover genus is 3, by #E(K) mod 4; n under each group is the number of curves. Derived as a ratio of two recorded counts. q = 11 is a sample of 120.

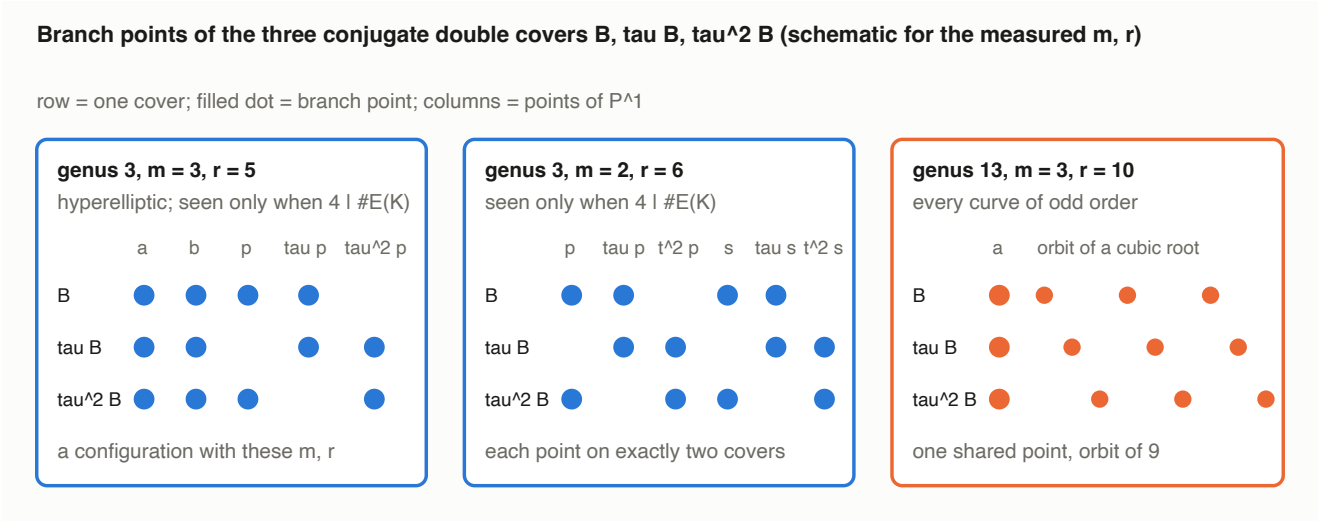


Figure 2: Schematic of the branch configurations behind the measured (m, r) types.

Missing data

- Isogeny-class minimum at q = 11 (sample only).
- Hyperellipticity of the m = 2 genus-3 covers: not decided by branch data.